

Automatic question generation from a given passage and an answer: improved generalization with unknown words using a knowledge distillation module

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1 Problem Overview

Automatic Question Generation (AQG)[1] is the process of creating questions from a given text or context, along with relevant answers. This process ensures contextually relevant questions and is valuable for generating questions for educational materials[2], enhancing machine reading comprehension systems[3], building web answering systems[4], conversational dialog systems[5] and chatbots[6].

Early AQG research primarily focused on producing grammatically correct and answerable questions[7, 8]. However, these early methods had limitations, as they didn't generate questions specific to answers in the passage. It was necessary for generating natural questions, considering multiple potential answers within the passage. The earlier methods also lacked continuity in the conversational tasks where questions needed to be contextually relevant to previous interactions[9].

Therefore, recent AQG models[10, 11, 12, 13] have incorporated the target answer information into question generation, employing complex networks to encode passages, answers, and their correlations, followed by question generation. However, these approaches face two primary limitations. First, many of these methods generate questions in a single pass, without explicitly modeling the correlation between the target answer and the question. This can lead to questions that are grammatically correct but not answerable. Second, there has been limited research on generating questions with new or uncommon words. These less common words enhance question diversity and improve question quality in terms of naturalness and answerability. In this work, I try to focus on the integration of question generation with question answering to ensure the consistency of the generated questions, and the generation of questions containing uncommon words for better generalization.

2 Relevancy to the class

My project is directly related to the topics covered in my NLP class. It will utilize word embedding and sequential embedding techniques to create coherent and contextually relevant embedding of given texts. The framework and its attention mechanism are also inspired by the structure of transformers. Moreover, it is using huggingface library and pre-trained BART model which were taught in our Large Language Models (LLM) lecture.

3 Project Description

The architecture I have designed, unifies three major tasks: Question Generation, Question Answering and Unknown Word Generation through Knowledge Distillation. If we observe the structure, we can see that, it has An embedding Layer, A contextual layer consists (stacked layers of Autoencoders), A stacked layer of Decoder, A pretrained model for knowledge distillation and output layers for each tasks. The complete architecture is given in Fig: 1.

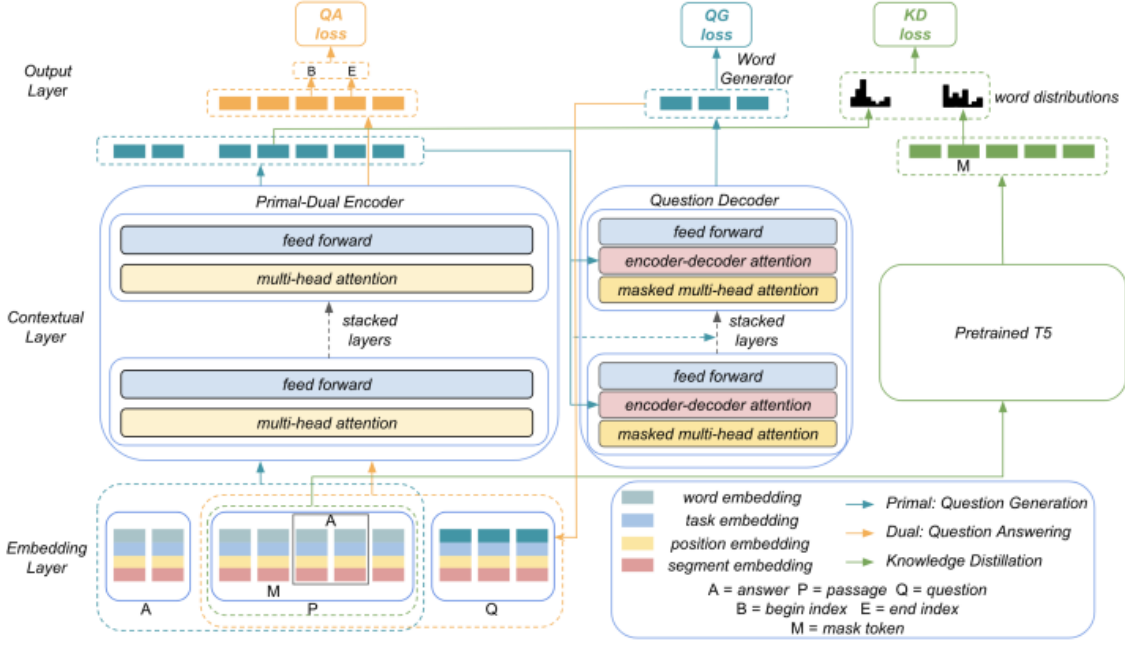


Figure 1: Structure Of The model

3.1 Structural Description

3.1.1 Embedding Layer:

The embedding layer of is consist of four embedding units. Word embedding, position embedding, task embedding, segment embedding. The word embedding and the position embedding is self-explanatory. The task embedding basically denotes, for which task I am embedding the input. task=0 for Question Generation, 1 for Question Answering and 2 for Uncommon Word generation. In the tasks, for some operations, we have to concatenate two input pairs: such as passage-answer pair or passage-question pair. Therefore, Therefore segment embedding is used to determine the origin of segments in the input. Segment ID = 0 for context, 1 for answer and 2 for question.

3.1.2 Primal Dual Encoder:

This is a simple encoding layer like general transformer architecture. However, this single encoder is being used for both the tasks: Question Generation and Question Answering. Therefore it is being called as primal-dual encoder. The whole encoder is a stacked layers of encoder units where each unit consist of a multi-head attention and a feed-forward layer. In my implementation, I have used a pretrained BART encoder, which has such six stacked units. The layer receives embedded input from the embedding layer and produces encoded outputs in the last hidden layer.

3.1.3 Question Decoder:

This is a transformer-like decoder used for question-generation task. In my implementation, I used a pretrained BART decoder which is consist of stacked layers of six decoding unit. Each unit consist of similar sub-layers like encoder with an extra encoder-decoder attention layer. It takes the last state of encoder, and sample question shape as input and produces output the generated question. Here, encoder-decoder attention layer helps to tune the encoder through the connection with last layer of encoder taken as input.

3.1.4 Output Layers:

There are three output layers dedicated to each task and they produce outputs of shapes desired.

3.2 Task Description

3.2.1 Question Generation

The input for question generation is the concatenation of the passage and the answer. The input is first embedded and then passed through the encoder. The encoder produces encoded output which is later passed to the decoder as input. The decoder produces its decoded output. Then I used a pre-trained BART sequence to sequence converter as the output layer. In the output, a sequence is generated which is the desired question of concatenated input of passage and answer pair.

3.2.2 Question Answering

Though the final output target of this model is question generation, it has question-answering subpart based on the same encoder user earlier for question generation. The target is to tune the encoder to that perfection, so that it can learn the pattern of the question suitable for answering the task perfectly. As the encoder is being used for both the tasks, it is being called the primal-dual encoder. For this task, passage and the question are used as input. after embedding, it is passed through the same encoder. From the output of the encoder, an output layer is used to determine the start and the end index of the answer. This means, from what position in the passage, the answer starts and where it ends.

3.2.3 Unknown Word Generation

This task works by generalizing the model to a pretrained language model. It basically compares the output between if the passage were fed to a pre-trained model vs if the message is fed to the encoder. The motivation is to generalize the model by tuning its encoder for generating words out of the passage.

3.3 Generating the Question

We might have observed that the decoder is taking the question as input as well during the training phase. But when we are to finally generating the question, what will be in the place of decoder input? It is basically done by using beam search. At first, in place of the question, only a dummy hypothesis consisting starting and end token is passed. Based on the passage and answering input, it generates the first word. Then through beam search, top k possible first word is selected and by adopting predicted first word, the process proceeds to generate the second word. Through this iterative process, it finally able to generate the question.

3.4 Experimentation Details

The model was trained and evaluated on the widely used benchmark Stanford Question Answering Dataset (SQuAD)[14] dataset. The original SQuAD dataset contains 23215 paragraphs from 536 Wikipedia articles with over 100k questions posed about the articles. I have used BLEU (Bilingual Evaluation Understudy) as the evaluation metric. The following loss function has been used for evaluation:

$$L_{total} = L_{QG} + \alpha * L_{QA} + \beta * L_{UW} \quad (1)$$

Here, α and β denote the contribution of loss in question answering and unknown word generation, respectively. According to [9], I have used $\alpha = 0.8$ and $\beta = 0.15$.

3.5 Results and Comparisons

My implementation achieved a BLEU score of 1.275913 where as the original paper [9] achieved a BLEU score of 19.07.

4 Ideas from the class I have used

The structure of this model is based on the basic transformer structure. Therefore, the idea of transformer, the breakdown of each component and layers such as encoder, decoder, attention mechanism

and mutual attention has been directly used in this project. The project also used a multi-layer embedding. In this task, I have used the idea of embedding from the class. And finally, the project uses large language model BART and the huggingface library. Which was also covered in the class.

5 What have learned

This project was a great journey of learning for me. I have incorporated all the ideas that I learnt from the class. Moreover, during the project: I learnt different implementation details. For example: while I was managing a separate position input tensor for the embedding task, I came to learn that the word embedding and the positional embedding is coupled at the embedding layer of BART. Therefore, no additional positional embedding is needed.

I also have learned how to use pretrained model components for implementing the task. I have used many components of pretrained BART model for the task.

I also learned many implementation nuances, such as, using the ending token in the beam-search hypothesis can lead the model to produce end of the question token.

Above all, the question generation task was new to me and I have learnt how to implement it.

6 If I had much more time, how would I continue the project

Firstly, I would have solved the major data-leaking problem in the question answering task that I identified at the eleventh hour. It is causing severe degradation in performance and my performance is standing faraway from the referenced paper.

I drew inspiration for my proposed solution from a paper presented at EMNLP-2022, titled "Learning to Generate Questions by Posing Questions: An Innovative Primal-Dual Approach for Generating Uncommon Words[9]." If I had much time, I would have explored the influence of knowledge distillation in unknown word generation in depth.

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